AI-Based Gender Stereotype Detection and Educational Content Improvement

SDG 5 – Gender Equality

1. Problem Statement

Educational books and learning materials can contain gender stereotypes through the way men and women are represented in occupations, family responsibilities, activities, personality traits, and visual content. Repeated exposure to such patterns may contribute to students developing traditional assumptions about gender roles. However, manually reviewing large volumes of educational content is time-consuming and subjective. The proposed system aims to develop an AI-assisted platform that detects potential gender stereotypes in educational content, provides evidence for the detected pattern, and uses agentic AI to suggest context-appropriate revisions while preserving the original educational objective.

2. Abstract

2.1 Introduction

This project proposes an AI-based system for identifying and improving potential gender stereotypes in educational textbooks and related learning materials. The system analyzes text and, optionally, textbook images to identify patterns such as unequal gender representation, traditional occupation associations, and stereotypical domestic or social roles. Rather than making a simple binary judgment, the system presents evidence, context, and confidence for each potential issue. An agentic workflow is then used to investigate the finding, retrieve relevant supporting knowledge, generate an alternative version of the content, and review the proposed change.

2.2 Design & Implementation

The system is designed as a modular pipeline. First, textbook PDFs are processed to extract text and images. The extracted text is divided into manageable sections while preserving page and chapter information. A machine-learning-based stereotype detection model analyzes the text and identifies potential stereotype categories such as gender-role association, occupation association, representation imbalance, and stereotypical descriptions.

The detected cases are passed to an agentic workflow implemented using LangGraph. An Investigator Agent examines the surrounding context and the detector's evidence to determine whether the pattern is meaningful. A retrieval component can provide research literature, educational guidelines, and the project's stereotype taxonomy as supporting context. A Recommendation Agent, using a capable frontier language model, proposes a revised passage that aims to reduce the stereotype without unnecessarily

changing facts or the intended learning objective. A Reviewer Agent checks the proposed revision for meaning preservation, age appropriateness, factual consistency, and whether it introduces another stereotype. Rejected suggestions can be sent through the workflow again for revision.

The final result can be presented through a web dashboard showing the original passage, detected pattern, supporting evidence, confidence, suggested revision, and review status. For images, a vision-language model can optionally identify gender representation and role patterns, allowing text and visual evidence to be considered together.

2.3 Future Enhancement

Future versions can support multiple Indian languages using multilingual models, OCR for scanned textbooks, and stronger computer-vision analysis of illustrations. The platform can also include teacher/editor approval workflows, version comparison, explainable bias analytics, curriculum-specific rules, and longitudinal analysis across different editions of a textbook. A human-in-the-loop review system can ensure that AI-generated suggestions remain recommendations rather than automatic replacements. The project can also be extended to other forms of educational representation, including disability, socioeconomic representation, and cultural stereotypes.

3. Brief Review

Existing research shows that gender representation in educational materials can be uneven and that textbooks may associate men and women with different occupations, activities, and social roles. Research on school textbooks has used qualitative content analysis, statistical analysis, and natural language processing to measure representation and gender-associated language. Recent large-scale work has also demonstrated that automated text analysis can be used to study gender bias across textbooks from multiple countries.

Research focused on India has examined both textual and visual representation in school textbooks, including the distribution of male and female references and the roles shown in textbook illustrations. These studies provide a useful foundation for defining stereotype categories and measurable indicators. The proposed project extends this direction by combining an explicit detection model with an agentic workflow that explains findings and generates and reviews possible content improvements.

4. Tools and Technologies Used

- **Python:** Data processing, NLP pipeline, model development, and evaluation.
- **PyMuPDF / PDF parser:** Extraction of text, pages, and document structure from textbooks.
- **OCR:** Extraction of text from scanned or image-based textbook pages when required.

- **BERT / RoBERTa / DeBERTa or IndicBERT:** Candidate models for fine-tuning the stereotype detection classifier.
- **LangChain:** LLM integration, prompts, structured outputs, retrieval, and tool integration.
- **LangGraph:** Agent orchestration, state management, conditional routing, loops, and review workflow.
- **Frontier Language Model:** Context analysis, recommendation/rewrite generation, and review.
- **Vision-Language Model:** Optional analysis of textbook images and visual gender-role representation.
- **RAG + Vector Database:** Retrieval of research papers, educational guidelines, and project-specific knowledge.
- **FastAPI:** Backend APIs for processing requests and connecting the AI pipeline to the application.
- **React.js:** Interactive frontend/dashboard for displaying analysis and recommendations.
- **PostgreSQL:** Storage of documents, analysis results, evidence, recommendations, and review status.

5. Applications

- Textbook auditing: Assist publishers, curriculum teams, and educational institutions in reviewing large volumes of educational material.
- Content improvement: Provide evidence-backed suggestions for making examples and descriptions more balanced and inclusive.
- Teacher and editor assistance: Help educators identify potentially stereotypical passages before publication or classroom use.
- Research and analytics: Measure gender representation across chapters, subjects, books, editions, or educational levels.
- Educational policy support: Generate structured evidence that can support reviews of learning materials against gender-equality objectives.
- Multimedia content auditing: Extend analysis from text to textbook illustrations and other educational visual material.
- Awareness and training: Demonstrate common forms of gender stereotyping and help content creators understand how wording and examples can influence representation.

6. SDGs Used

6.1 Primary SDG – SDG 5: Gender Equality

SDG 5 aims to achieve gender equality and empower all women and girls. This project directly supports the goal by addressing gender representation in educational content and providing tools to identify and reduce potentially stereotypical portrayals.

6.2 Related SDGs

- **SDG 4 – Quality Education:** Supports inclusive and equitable educational content and contributes to improving the quality of learning materials.
- **SDG 10 – Reduced Inequalities:** Can help identify representational inequalities and support more inclusive educational resources.

7. Proposed System Outcome

The expected output is an AI-assisted educational content auditing platform. For every flagged case, the platform should provide: (1) original content, (2) detected stereotype category, (3) supporting evidence and context, (4) model confidence, (5) suggested revision, and (6) reviewer feedback/status. The system is intended to assist human editors and educators, not to automatically replace their judgment.